

Cycles in nature

NUTRIENTS AND OTHER SUBSTANCES ARE RECYCLED IN THE ENVIRONMENT.

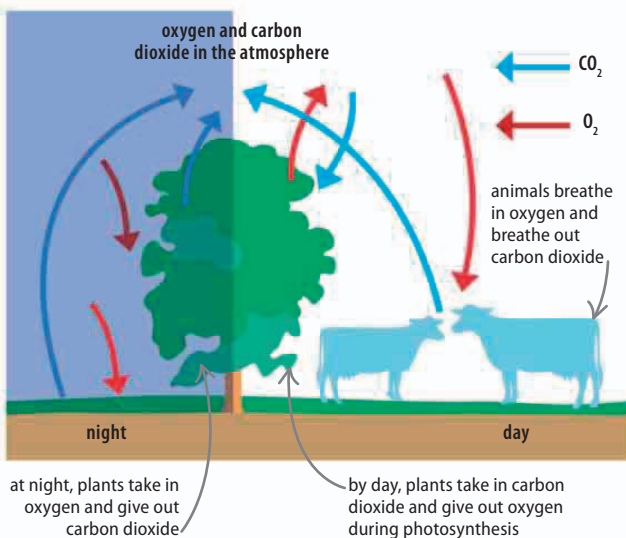
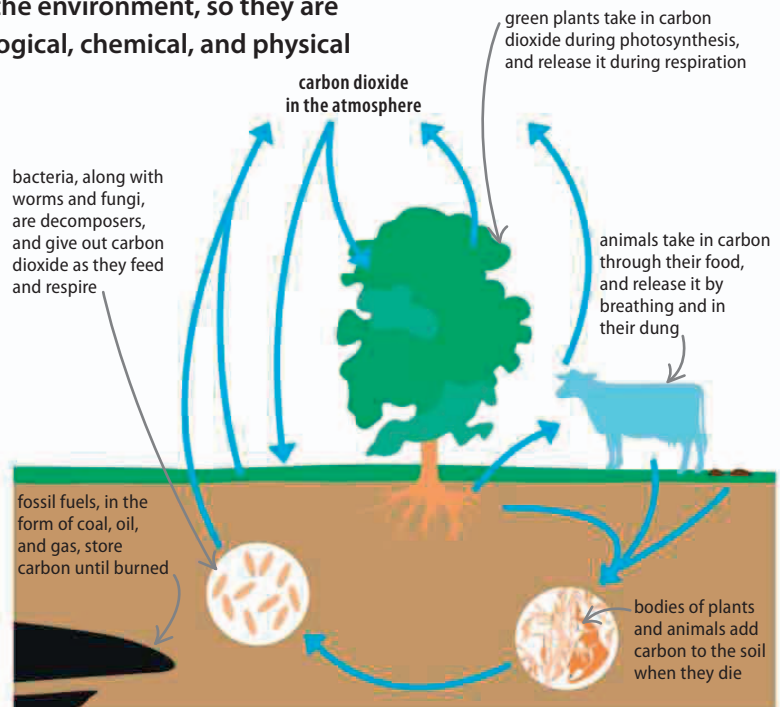
Living things require many nutrients—substances used to build their bodies. There is a finite supply of these in the environment, so they are recycled through the environment by biological, chemical, and physical processes—and also by human activities.

The carbon cycle

Carbon is essential to life. It is one of the most abundant elements in a living body and its atoms are in just about every chemical in cells. During photosynthesis, plants fix (collect) carbon dioxide from the atmosphere and turn it into sugars and other nutrients. These then pass to animals and other organisms that eat the plants. Eventually, the carbon in them is returned to the atmosphere as a waste product of respiration.

▷ Nonbiological factors

Carbon is not only cycled between the atmosphere and organisms. Carbonates, a combination of carbon and oxygen found in rocks and fossil fuels, are locked away underground for millions of years. Burning fossil fuels releases this carbon dioxide back into the atmosphere.



The oxygen cycle

Almost all organisms require a supply of oxygen, which is used in respiration to release energy from sugar. Organisms take in the oxygen and give out carbon dioxide (a waste product of respiration). However, oxygen does not run out, because it is constantly being replaced by the photosynthesis of plants. In this process carbon dioxide is taken in as a raw ingredient of glucose, and oxygen is given out as a waste material.

◁ Night and day

Plants take in carbon dioxide as a raw material for photosynthesis, and give out oxygen as a waste material of the process. Plants photosynthesize only during daylight, and this is when oxygen is released into the atmosphere. By night, plants take in some oxygen to power their respiration, but they use less than they produce.

SEE ALSO

◁ 28–29 Respiration

◁ 30–31 Photosynthesis

◁ 34–35 Waste materials

Chemical industry 154–155 ▷

Carbon and fossil fuels 156–157 ▷